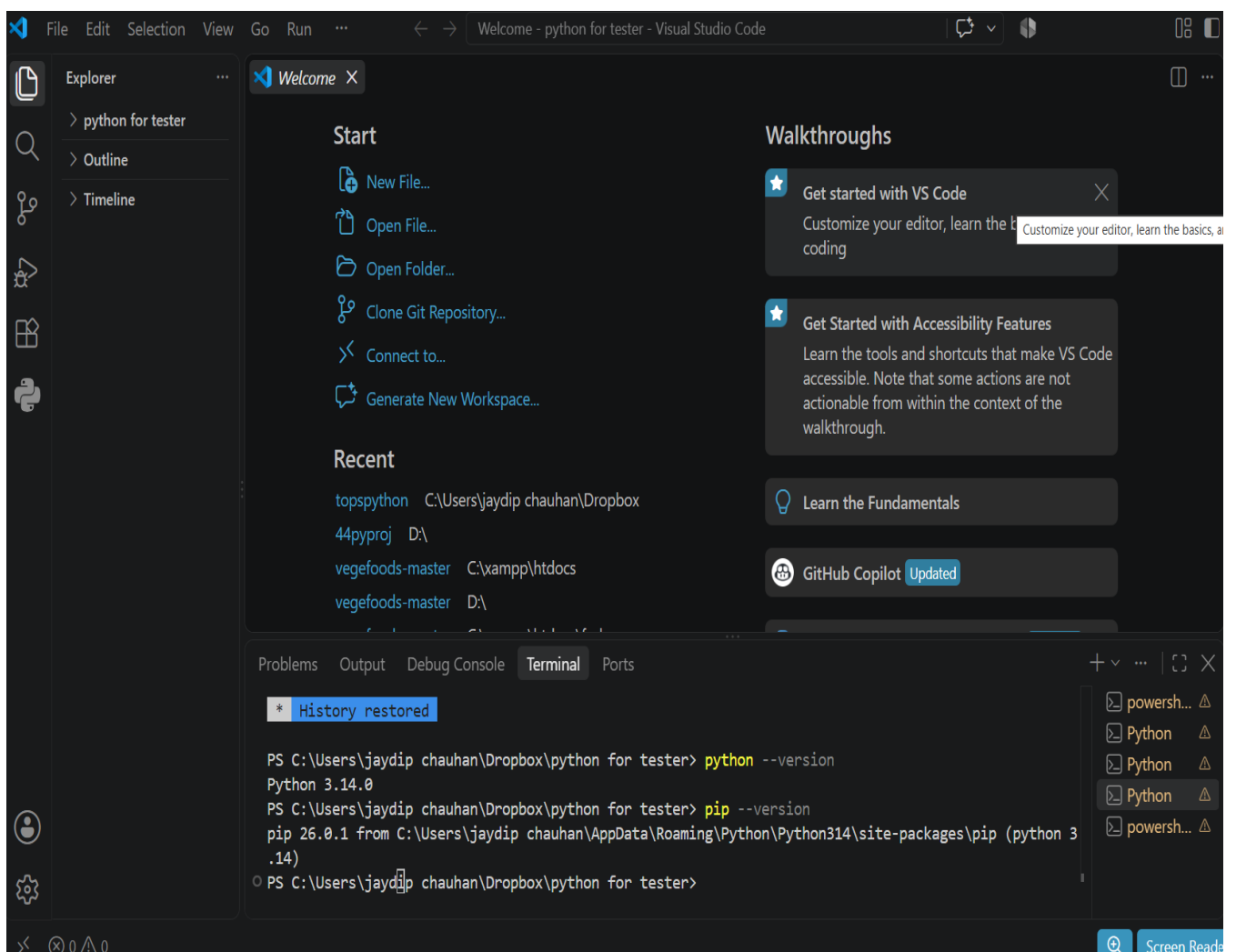


SESSION 1: INTRODUCTION TO PYTHON & ENVIRONMENT SETUP

TASKS:

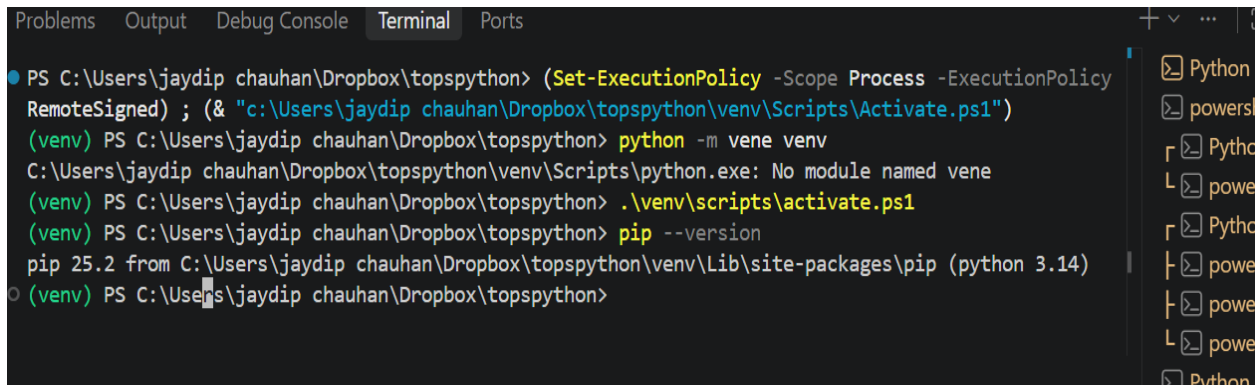
1. Install Python (latest version) and VS Code on your system, then take a screenshot of both the Python installation confirmation and the VS Code welcome screen.

Ans:



2. Create a new folder named Inst Test Project, open it in VS Code, and set up a Python virtual environment inside it using the venv module.

Ans: I created a new folder named **InstaTestProject** and opened it in VS Code. Then I created a Python virtual environment using the venv module.



```
Problems Output Debug Console Terminal Ports
● PS C:\Users\jaydip chauhan\Dropbox\topspython> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\jaydip chauhan\Dropbox\topspython\venv\Scripts\Activate.ps1")
(venv) PS C:\Users\jaydip chauhan\Dropbox\topspython> python -m venv venv
C:\Users\jaydip chauhan\Dropbox\topspython\venv\Scripts\python.exe: No module named venv
(venv) PS C:\Users\jaydip chauhan\Dropbox\topspython> .\venv\scripts\activate.ps1
(venv) PS C:\Users\jaydip chauhan\Dropbox\topspython> pip --version
pip 25.2 from C:\Users\jaydip chauhan\Dropbox\topspython\venv\Lib\site-packages\pip (python 3.14)
○ (venv) PS C:\Users\jaydip chauhan\Dropbox\topspython>
```

3. Use pip inside your InstaTestProject virtual environment to install the requests package, then run a 'pip list' and note the installed packages.

Ans:

```

PS C:\Users\python -m pip install --upgrade pip
Requirement already satisfied: pip in C:\Users\jaydip chauhan\AppData\Roaming\Python\Python314\site-packages (26.0.1)
Collecting pip
  Using cached pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.2.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 26.0.1
    Uninstalling pip-26.0.1:
      Successfully uninstalled pip-26.0.1
  WARNING: The scripts pip.exe, pip3.14.exe and pip3.exe are installed in 'C:\Users\jaydip chauhan\AppData\Roaming\Python\Python314\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-26.2.1
PS C:\Users\jaydip chauhan\Dropbox\assignment> python -m pip show requests
Name: requests
Version: 2.34.2
Summary: Python HTTP for Humans.
Home-page:
Author:
Author-email: Kenneth Reitz <me@kennethreitz.org>
License: Apache-2.0
Location: C:\Users\jaydip chauhan\AppData\Roaming\Python\Python314\site-packages
Requires: certifi, charset_normalizer, idna, urllib3
Required-by:
PS C:\Users\jaydip chauhan\Dropbox\assignment> pip list
Package            Version
-----
certifi            2026.7.22
charset-normalizer 3.5.1
colorama           0.4.6

```

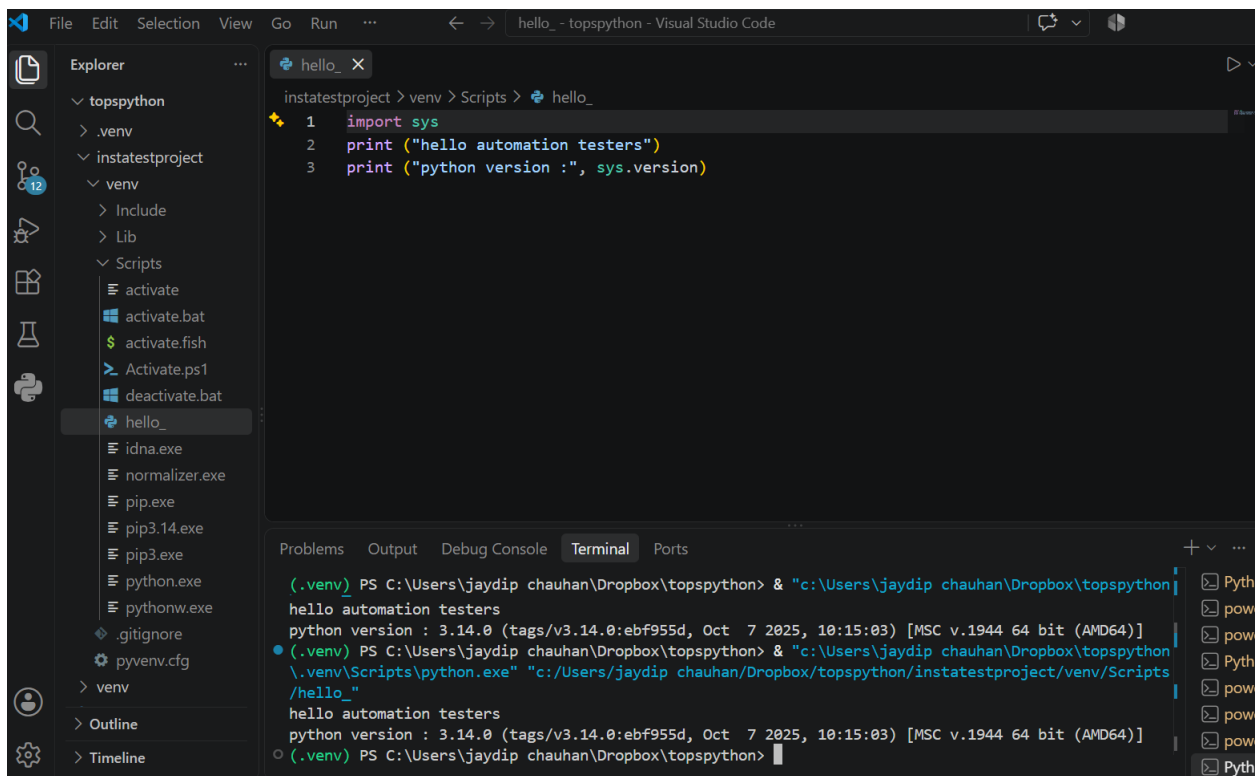
```
Problems Output Debug Console Terminal Ports
Location: C:\Users\jaydip chauhan\AppData\Roaming\Python\Python314\site-packages
Requires: certifi, charset_normalizer, idna, urllib3
Required-by:
PS C:\Users\jaydip chauhan\Dropbox\assignment> pip list
Package Version
-----
certifi 2026.7.22
charset-normalizer 3.5.1
colorama 0.4.6
contourpy 1.3.3
cycler 0.12.1
fonttools 4.62.1
idna 3.19
joblib 1.5.3
kiwisolver 1.5.0
matplotlib 3.10.8
numpy 2.4.3
packaging 26.0
pandas 3.0.1
pillow 12.1.1
pip 26.2.1
pyparsing 3.3.2
python-dateutil 2.9.0.post0
qrcode 8.2
requests 2.34.2
scikit-learn 1.8.0
scipy 1.17.1
seaborn 0.13.2
six 1.17.0
threadpoolctl 3.6.0
tzdata 2025.3
urllib3 2.7.0
```

4. Write and run a Python script named `hello_automation.py` that prints Hello, Automation Tester! and the current Python version is being used.

Hint: Use the `sys.version` property to display the Python version.

Ans: I created a Python file named **hello_automation.py** and added the following code:

The program displays the message and the current Python version being used.



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left displays the file structure of a project named 'topspython', including a virtual environment (venv) and its Scripts directory. The main editor window shows a file named 'hello_'. The code in the editor is as follows:

```
1 import sys
2 print("hello automation testers")
3 print("python version :", sys.version)
```

The Terminal panel at the bottom shows the command prompt output after running the script. The output is:

```
(.venv) PS C:\Users\jaydip chauhan\Dropbox\topspython> & "c:\Users\jaydip chauhan\Dropbox\topspython\venv\Scripts\python.exe" "c:\Users\jaydip chauhan\Dropbox\topspython\instatetestproject\venv\Scripts\hello_"
hello automation testers
python version : 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
(.venv) PS C:\Users\jaydip chauhan\Dropbox\topspython>
```

5.Explain in 3-4 lines how using a virtual environment (venv) helps when working on multiple automation projects, especially if you need different package versions for each.

Ans:

A virtual environment keeps each automation project separate from other projects.

It allows us to install different package versions for different projects.

Changes in one project do not affect another project.

This makes Python automation projects easier to manage and maintain.